

```

import cv2
import matplotlib.pyplot as plt
from google.colab import files

# Upload surveillance image
print("Upload an image containing people or vehicles")
uploaded = files.upload()

filename = list(uploaded.keys())[0]

# Read image
img = cv2.imread(filename)
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# -----
# Image Preprocessing
# -----
gray = cv2.equalizeHist(gray)

# -----
# Object Detection (Face Detection Example)
# -----
face_cascade = cv2.CascadeClassifier(
    cv2.data.haarcascades + "haarcascade_frontalface_default.xml"
)

faces = face_cascade.detectMultiScale(
    gray,
    scaleFactor=1.1,
    minNeighbors=5,
    minSize=(30,30)
)

# Draw bounding boxes
output = img_rgb.copy()

for (x, y, w, h) in faces:
    cv2.rectangle(output, (x, y), (x+w, y+h), (0,255,0), 2)

# -----
# Display
# -----

plt.figure(figsize=(12,6))

plt.subplot(1,2,1)
plt.imshow(img_rgb)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(output)
plt.title("Detected Faces")
plt.axis("off")

plt.show()

# -----
# Features of Computer Vision
# -----

print("Computer Vision Features in Automated Surveillance")
print("-----")
print("1. Image Acquisition")

```

```
print("2. Image Enhancement")
print("3. Object Detection")
print("4. Object Tracking")
print("5. Face Recognition")
print("6. Decision Making and Alert Generation")

print(f"\nNumber of Faces Detected: {len(faces)}")
```

Upload an image containing people or vehicles

WhatsApp Image 2026-01-09 at 11.18.59 AM.jpeg

WhatsApp Image 2026-01-09 at 11.18.59 AM.jpeg(image/jpeg) - 271415 bytes, last modified: 17/01/2026 - 100% done
Saving WhatsApp Image 2026-01-09 at 11.18.59 AM.jpeg to WhatsApp Image 2026-01-09 at 11.18.59 AM.j

Original Image



Detected Faces



Computer Vision Features in Automated Surveillance

-
1. Image Acquisition
 2. Image Enhancement
 3. Object Detection
 4. Object Tracking
 5. Face Recognition
 6. Decision Making and Alert Generation

Number of Faces Detected: 3

